SYSTEM ARCHITECTURE

Credit Risk Portfolio Analytics | Anshul Pundhir (anshulpundhirwork@gmail.com)

1. End-to-End Pipeline Flow (Raw Data to Executive Dashboard)

Stage & Script	Libraries & Tools	Data Flow, Operations & Intentional Quality Treatments
1. Raw Ingestion & QC `src/ingestion.py`	Python 3.12 Pandas NumPy PyArrow	**Inputs:** 5 raw CSVs (customers, loans, bureau_credit, repayment_history, collections). **Operations:** Standardizes string labels (e.g., gender), applies caps to anomalous numeric bounds (tenure exceeding age), and imputes missing interest rates via loan-type medians. Resolves missing employment data using cohort medians, inserting strict `_is_na` audit flags for regulatory tracing. Outputs validated Parquet files.

2. Feature Store
`src/features.py`
`src/analytics.py`
	Scikit-Learn Pandas	**Operations:** Aggregates external bureau trade-lines to borrower grain (utilization, total lines, write-offs). Computes monthly payment roll-rates and trailing 3-month momentum. Calculates annualized DTI and principal LTI. **Critical Step:** Strictly filters all post-default collections events to prevent target data leakage into the training set.

3. ML & Anomaly Engine
`src/model.py`
`src/anomaly.py`
	LightGBM SHAP SciPy Joblib	**Operations:** Trains an interpretable Logistic Regression benchmark alongside a depth-constrained LightGBM champion model. Pre-computes per-account **TreeSHAP** local attribution vectors. Runs a dual-lens unsupervised anomaly engine (Lens 1: Trailing payment drop heuristics; Lens 2: Cohort-normalized Isolation Forest) to flag pre-bureau risk outliers.

4. Dashboard (UI)
`app/app.py`
`app/views/*`
	Streamlit Plotly	**Operations:** Interactive interface featuring Portfolio Overview (macro KPIs, multidimensional heatmaps), Account Deep-Dive (gauge charts, SHAP waterfall, repayment timelines), Anomaly Triage Queue, and the LLM-powered natural language Copilot.

2. Embedded AI Systems Architecture & Data Boundaries

AI Feature	Trigger & Module	Mechanics, LLM Calls & Data Payload (Privacy Shield)
Ask the Portfolio (Natural Language Q&A)	**Trigger:** Analyst submits query in UI. **Module:** `src/llm_assistant.py`	**Mechanism:** The query is intercepted and its intent is mapped. A deterministic Pandas backend computes the exact answer using the filtered dashboard state. The LLM is called strictly to format the response into conversational prose. **LLM Call:** `Google Gemini API` or `OpenAI API`. **Payload Sent:** ONLY a JSON block containing pre-aggregated metrics (cohort default rates, dollar exposure, total n-count, and calculation formula). **Privacy Shield:** ZERO raw customer PII (no account IDs, names, or transaction rows) is ever transmitted across the external API network boundary.
Anomaly Detection (Dual-Lens Outliers)	**Trigger:** Batch run on data refresh. **Module:** `src/anomaly.py`	**Mechanism:** Entirely localized, deterministic, and offline (No LLM network call). **Lens 1:** Tracks 3-month payment velocity drop. **Lens 2:** `sklearn.ensemble.IsolationForest` identifies multidimensional peer outliers based on cohort-normalized DTI, LTI, and bureau utilization matrices. Flags accounts scoring >= 35.